COURSE : Data Analytics

SQL Aggregate functions & operators

SQL Foundations : Excercise 02

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01 Students Table

1. Distinct departments

```
SELECT DISTINCT departments
FROM Students;
```

department
IT
HR
Finance

2. Average age per department

```
SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department;
```

department	avg_age
IT	20.5
HR	22
Finance	23

```sql
SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) >1;
```

department	student_count
IT	2
HR	2

4. Age between 21 and 23

```sql
SELECT
FROM
WHERE age BETWEEN 21 AND 23;
```

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. IT or HR and age >21

```sql
SELECT studen_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR')
AND age >21;
```

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

02 COURSES table

6. Total Credits per department >5

```
SELECT department, SUM(Credits) AS total_credits
FROM Courses
GROUP BY department
HAVING SUM (Credits)>5
```

department	total_credits
IT	11

7. Courses not equal to 4 Credits

```
SELECT Course_id, Course_name, department, Credits
FROM Courses
WHERE Credits != 4
```

Course_id	Cours_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Top 3 Courses by Credits

```
SELECT Cours_id, Course_name, credits
FROM Courses
ORDER BY credits DESC
LIMIT 3;
```

Course_id	Course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

03 Enrollment Table

9. Max, Min, Avg grade

```
SELECT
MAX (grade) AS Max_grade
MIN (grade) AS Min_grade
AVG (grade) AS avg_grade
FROM enrollments
```

Max_grade	Min_grade	avg_grade
90	78	84.6

10. Count enrollments per course.

```
SELECT course_id, COUNT (*) AS enrollment_count
FROM enrolments
GROUPBY Course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

04 Salaries Table

11. Total Salary & bonus per department

```
SELECT deparment
SUM (Salary AS total_Salary
SUM (bonus) AS total_bonus
FROM Salaries
GROUPBY department;
```

department	total_salary	total_bonus
IT	122 000	10500
HR	109 000	7500
Finances	70 000	6000

2. Avg Salary > 55000

```
SELECT department, AVG (Salary) AS avg_salary
FROM Salaries
GROUP BY department
HAVING AVG (Salary > 55000;
```

department	avg_salary
IT	61 000
Finance	70 000

3. Salary + bonus > 60000.

```
SELECT employee_id, name, salary, bonus, (Salary+bonus)
AS total Compensation
FROM Salaries
WHERE (Salary + bonus ) > 60 000
```

Employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5500	67 500

05 Projects Table

14. Total & avg budget > 70 000

SELECT

SUM (budget) AS total_budget

AVG (budget) AS avg_budget

FROM projects

GROUP BY departments

HAVING AVG (budget) > 70 000

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. Budget between 50k - 120k (exclude marketing)

SELECT projects_id, project_name, department, budget

FROM projects

WHERE budget BETWEEN 50 000 AND 120 000

AND department != 'Marketing';

Project_id	Project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	HR Portal	HR	50 000